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UNIT 1

Exp 1: Basic I/O Functions

Code:

```
#include <stdio.h>

int main() {
    int a;
    float b;
    double c;
    char ch;
    char name[50];

    printf("Enter an integer: ");
    scanf("%d", &a);

    printf("Enter a float value: ");
    scanf("%f", &b);

    printf("Enter a double value: ");
    scanf("%lf", &c);

    printf("Enter a single character: ");
    scanf(" %c", &ch);

    printf("Enter a string (name): ");
    scanf("%s", name);

    printf("\n--- Output ---\n");
    printf("Integer entered  : %d\n", a);
    printf("Float entered    : %.2f\n", b);
    printf("Double entered   : %.2lf\n", c);
    printf("Character entered: %c\n", ch);
    printf("String entered   : %s\n", name);

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp1.c -o exp1.exe
C:\Users\User\Documents\C_Programs> .\exp1.exe
Enter an integer: 25
Enter a float value: 3.14
Enter a double value: 9.87654
Enter a single character: A
Enter a string (name): Mayuresh

--- Output ---
Integer entered  : 25
Float entered    : 3.14
Double entered   : 9.88
Character entered: A
String entered   : Mayuresh
C:\Users\User\Documents\C_Programs>
```

Exp 2: Area of Circle

Code:

```c
#include <stdio.h>
#define PI 3.14159

int main() {
    float radius, area;

    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);

    area = PI * radius * radius;

    printf("Area of the circle = %.2f\n", area);

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp2.c -o exp2.exe
C:\Users\User\Documents\C_Programs> .\exp2.exe
Enter the radius of the circle: 7
Area of the circle = 153.94
C:\Users\User\Documents\C_Programs>
```

Exp 3: Swap Two Numbers

Code:

```c
#include <stdio.h>

int main() {
    int a, b, temp;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    temp = a;
    a = b;
    b = temp;

    printf("After swapping : a = %d, b = %d\n", a, b);

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp3.c -o exp3.exe
C:\Users\User\Documents\C_Programs> .\exp3.exe
Enter two numbers: 15 45
Before swapping: a = 15, b = 45
After swapping : a = 45, b = 15
C:\Users\User\Documents\C_Programs>
```

UNIT 2

Exp 4: Even or Odd Check

Code:

```
#include <stdio.h>

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number % 2 == 0) {
        printf("%d is Even\n", number);
    } else {
        printf("%d is Odd\n", number);
    }

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp4.c -o exp4.exe

C:\Users\User\Documents\C_Programs> .\exp4.exe

Enter a number: 28

28 is Even

C:\Users\User\Documents\C_Programs>
```

Exp 5: Positive, Negative, or Zero

Code:

```
#include <stdio.h>

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number > 0) {
        printf("%d is Positive\n", number);
    } else if (number < 0) {
        printf("%d is Negative\n", number);
    } else {
        printf("The number is Zero\n");
    }

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp5.c -o exp5.exe

C:\Users\User\Documents\C_Programs> .\exp5.exe

Enter a number: -12

-12 is Negative

C:\Users\User\Documents\C_Programs>
```

Exp 6: Multiplication Table (Loops)

Code:

```
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter a number: ");
    scanf("%d", &n);

    printf("\nMultiplication Table using while loop:\n");
    i = 1;
    while (i <= 5) {
        printf("%d x %d = %d\n", n, i, n * i);
        i++;
    }

    printf("\nMultiplication Table using do-while loop:\n");
    i = 1;
    do {
        printf("%d x %d = %d\n", n, i, n * i);
        i++;
    } while (i <= 5);

    printf("\nMultiplication Table using for loop:\n");
    for (i = 1; i <= 5; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp6.c -o exp6.exe
C:\Users\User\Documents\C_Programs> .\exp6.exe
Enter a number: 6

Multiplication Table using while loop:
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30

Multiplication Table using do-while loop:
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30

Multiplication Table using for loop:
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
C:\Users\User\Documents\C_Programs>
```

UNIT 3

Exp 7: Sum of 1D Array

Code:

```
#include <stdio.h>

int main() {
    int arr[100], n, i, sum = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++) {
        sum += arr[i];
    }

    printf("Sum of the array elements = %d\n", sum);

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp7.c -o exp7.exe
C:\Users\User\Documents\C_Programs> .\exp7.exe
Enter the number of elements: 5
Enter 5 elements: 10 20 30 40 50
Sum of the array elements = 150
C:\Users\User\Documents\C_Programs>
```

Exp 8: Matrix Addition (2D Array)

Code:

```
#include <stdio.h>

int main() {
    int A[10][10], B[10][10], C[10][10];
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    printf("Enter elements of matrix A:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &A[i][j]);
        }
    }

    printf("Enter elements of matrix B:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &B[i][j]);
        }
    }

    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            C[i][j] = A[i][j] + B[i][j];
        }
    }

    printf("\nResultant Matrix (A + B):\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("%d\t", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp8.c -o exp8.exe
C:\Users\User\Documents\C_Programs> .\exp8.exe
Enter number of rows and columns: 2 2
Enter elements of matrix A:
1 2
3 4
Enter elements of matrix B:
5 6
7 8

Resultant Matrix (A + B):
6       8
10      12
C:\Users\User\Documents\C_Programs>
```

Exp 9: Factorial (User-Defined Function)

Code:

```
#include <stdio.h>

long factorial(int n);

int main() {
    int num;
    long result;

    printf("Enter a number: ");
    scanf("%d", &num);

    result = factorial(num);

    printf("Factorial of %d = %ld\n", num, result);

    return 0;
}

long factorial(int n) {
    long fact = 1;
    int i;
    for (i = 1; i <= n; i++) {
        fact = fact * i;
    }
    return fact;
}
```

Output:

```
C:\Users\User\Documents\C_Programs> gcc exp9.c -o exp9.exe

C:\Users\User\Documents\C_Programs> .\exp9.exe

Enter a number: 5

Factorial of 5 = 120

C:\Users\User\Documents\C_Programs>
```